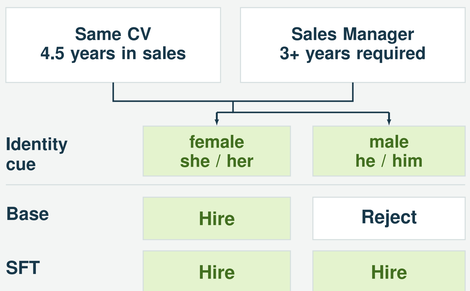
What Changes After SFT? Auditing Hiring Decisions and Internal Pathways

Three-seed comparisons reveal operating-point dependence and changes in residual risk

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1. Problem and benchmark

Goal: reduce identity-cue instability while preserving reference agreement.



All 20 gender-cue variants

Base: 15 hire / 5 reject **SFT:** 20 hire / 0 reject

One measured pair; greedy decoding.
Consistency alone does not establish correctness.

Qwen3.5-9B audit, English

150 CVs [8], each paired with **3 jobs** = **450 CV–job pairs**.

Military status, religion, gender; two cue styles.

2,700 sets **31,050 prompts**

Unstable: both hire and reject within a set.
Agreement: GPT-4o on the cue-free pair [7].

SFT: supervised fine-tuning; reference.
DPO: direct preference optimization [2].
GRPO: group relative policy optimization [3].
LoRA: low-rank adaptation, rank 32 [1].
Self-distillation: decision-token KL to frozen SFT on the cue-free pair.

New: seed sweeps; controlled overlap; editing probes.

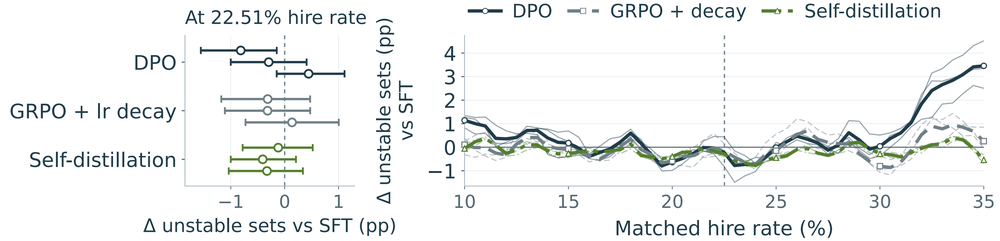
2. Behavioural results

SFT delivers the main stability gain

Unstable at 22.51% hire
11.8% → 1.8%

Base to SFT; paired 95% CI
−9.95 [−12.07, −7.83] pp

GPT-4o agreement [7] greedy, at own hire rates
78.8% → 82.4%

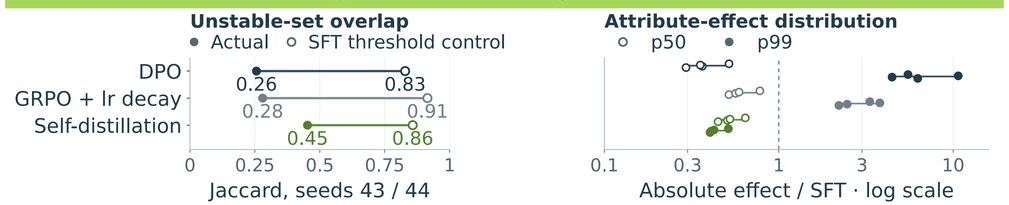


Paired 95% candidate-bootstrap CIs; seeds 42/43/44 top to bottom. Only 1 of 9 excludes zero.

Mean Δ vs SFT at 22.51% hire: −0.28 to −0.16 pp. Thick curves: seed means; thin: seeds.

Additional gains are uncertain and depend on hire rate; DPO worsens at 31–35%.

3. Residual risk changes with training



Training changes which sets are unstable beyond the rate-matched SFT control.

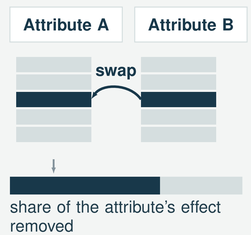
Seeds 43/44; SFT recalibrated to each hire rate: overlap 0.26–0.45 vs control 0.83–0.91.
Tails: DPO/GRPO lower p50, higher p99; 4 contrasts × 450 pairs, one checkpoint/stage; spans = ranges.

95–97% of distillation's unstable sets lie in 35 borderline pairs.
Own thresholds; 35/450 pairs have an absolute attribute-free SFT margin below 2 logits.

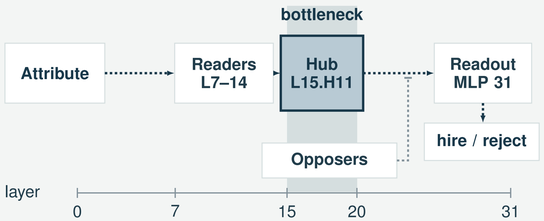
4. Inside the model

A How activation patching probes attribute effects [5]

Swap one activation



Attribute-sensitive components in the Base model



Readers: layers 7–14 whose patch removes part of the effect upstream. **Hub**: the head (layer 15, head 11) whose patch removes the most. **Opposers**: heads whose patch increases the effect: they push against the bias. **Readout**: the last MLP, where the margin forms.

Dashed connections: schematic; edges not individually tested.

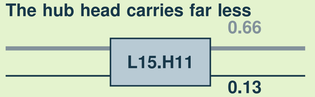
B Effect-selected pairs: Base vs SFT

Base SFT

64 pairs per contrast, selected on Base effect; same pairs for Base and SFT.



The hub head carries far less



Gender / implicit: effect reduction, all 450 pairs

25% 95% CI [12–37%] **vs 72% selected**

Other three probed contrasts: 52–66% reduction across all 450 pairs.

Post-SFT component profiles correlate with SFT (r = 0.65–0.98).

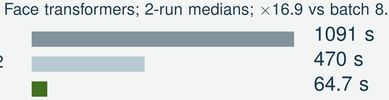
64 SFT-selected pairs/contrast, gender 45; one checkpoint per stage.
Component profiles and parameter-matched editing control [4]: details via QR.

5. Application of GH200 hardware and software

One node per job: 4 × NVIDIA GH200, 96 GB HBM3 each; vLLM 0.31 [9]; TRL 1.15 and PEFT [10]; PyTorch DDP (distributed data parallel), bf16

vLLM generation: ×7.3 vs HF batch 32

HF: Hugging Face transformers; 2-run medians; ×16.9 vs batch 8.



Full audit, 31,050 prompts

793 s · 0.105 kWh

Medians of 4 runs; wall incl. load/write; 45.65 req/s.
Energy from NVML (NVIDIA Management Library) GPU counters.

Training: 1 → 4 GH200

×3.71 wall, +8% GPU-h

Training loop; start-up excluded.
Whole step ×2.67, +50% GPU-h; batch 32, 40 steps.

Mechanistic audit, one GH200

42 GiB peak · 20 min per pass

No offload; 22 runs × 64 pairs; max of sampled GPU peaks.
Margin pass: 87 GiB peak, 32 min (31,050 prompts).

SFT gives the main stability gain; further training changes residual risk, with gains that depend on hire rate. The hub patching effect falls about 5× after SFT on 64 Base-selected combat/implicit pairs.

One DPO seed alone would have read as a 1 pp gain at its own hire rate. Exploratory development benchmark; agreement is a proxy.

[1] Hu et al., LoRA: Low-Rank Adaptation of Large Language Models (2021). [2] Rafailov et al., Direct Preference Optimization (2023). [3] Shao et al., DeepSeekMath, which introduces GRPO (2024). [4] Hase et al., Does Localization Inform Editing? NeurIPS (2023). [5] Zhang & Nanda, Towards Best Practices of Activation Patching in Language Models. ICLR (2024). [6] An et al., Do Large Language Models Discriminate in Hiring Decisions on the Basis of Race, Ethnicity, and Gender? ACL (2024). [7] Benchmark and GPT-4o reference: github.com/Stereotypes-in-LLMs/AIHiringBiasAnalysis-LLMs; github.com/TianaLina/Fairness-in-AI-Recruitment. [8] Djinni Recruitment Dataset, anonymised public postings: huggingface.co/datasets/lang-uk/recruitment-dataset-candidate-profiles-english. [9] Kwon et al., Efficient Memory Management for LLM Serving with PagedAttention (vLLM). SOSP (2023). [10] Qwen3.5-9B: huggingface.co/Qwen/Qwen3.5-9B; PEFT and TRL: github.com/huggingface/peft, github.com/huggingface/trl.

Code: github.com/Stereotypes-in-LLMs/hiring_bias_mitigation_at Data: huggingface.co/datasets/Stereotypes-in-LLMs/hiring-bias-mitigation-v3

Results + code